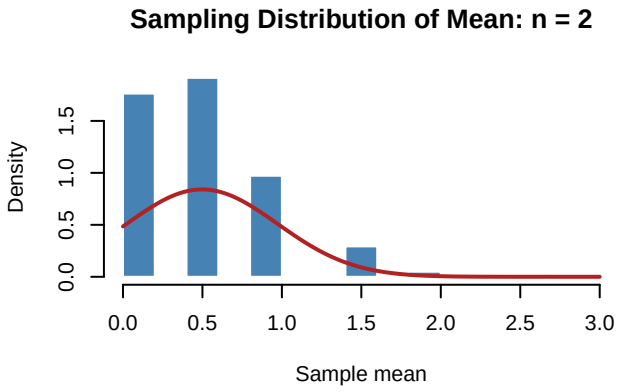
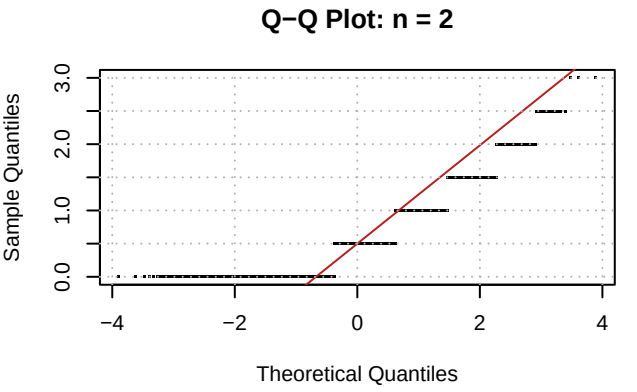
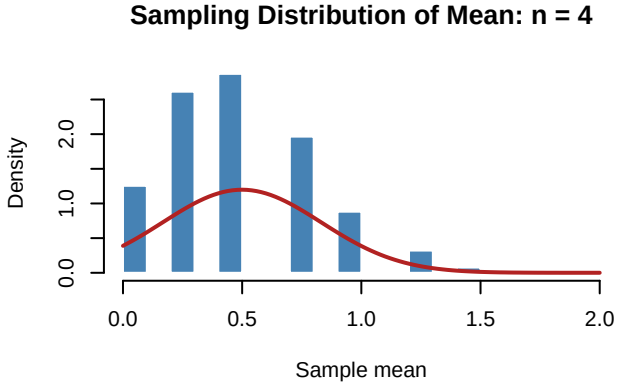
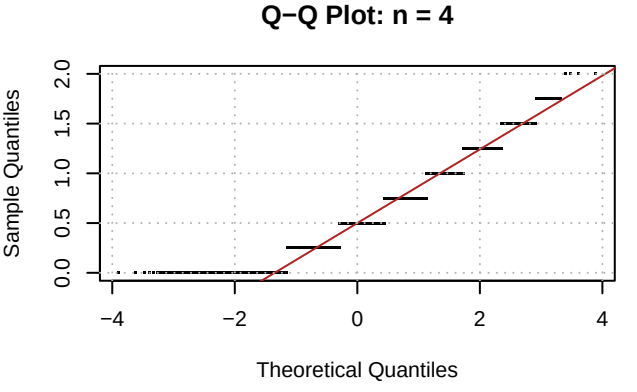
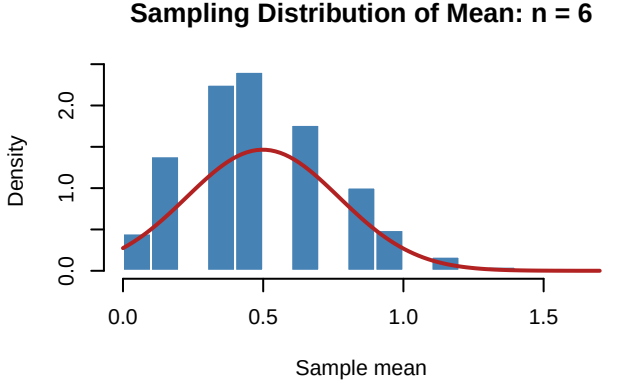
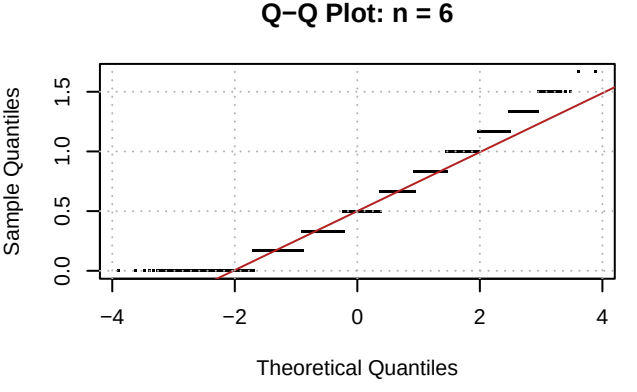

Sampling Distribution of Mean: n = 2
Density
0.0
0.5
1.0
1.5
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Sample mean

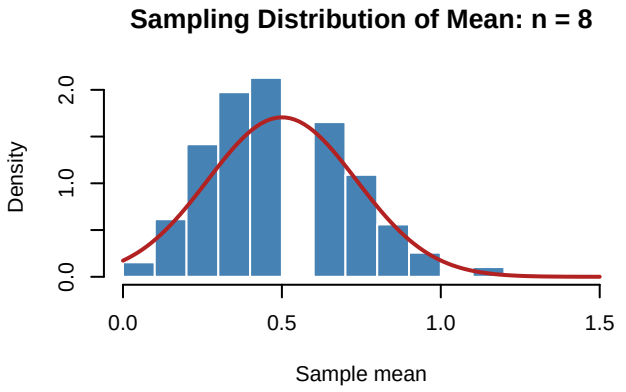

Q−Q Plot: n = 2
Sample Quantiles
0.0
1.0
2.0
3.0
−4
−2
0
2
4
Theoretical Quantiles

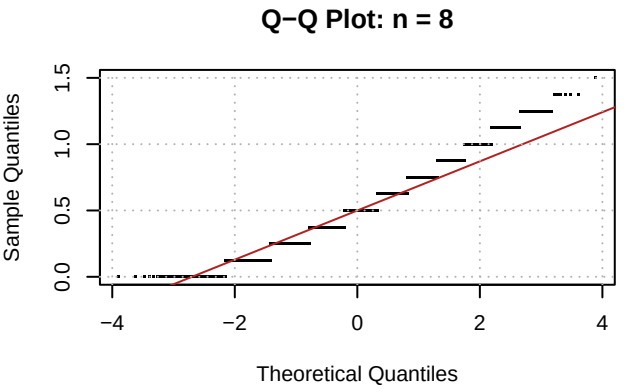

Sampling Distribution of Mean: n = 4
Density
0.0
1.0
2.0
0.0
0.5
1.0
1.5
2.0
Sample mean

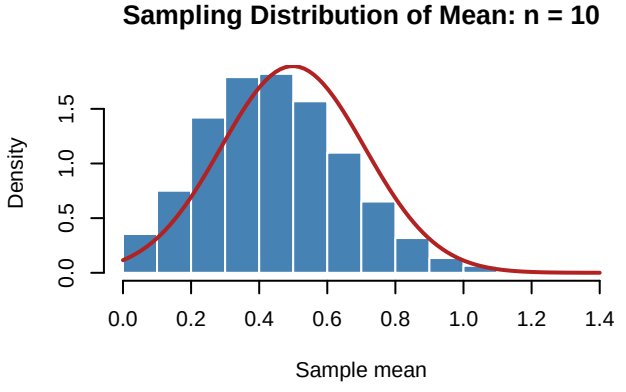

Q−Q Plot: n = 4
Sample Quantiles
0.0
0.5
1.0
1.5
2.0
−4
−2
0
2
4
Theoretical Quantiles

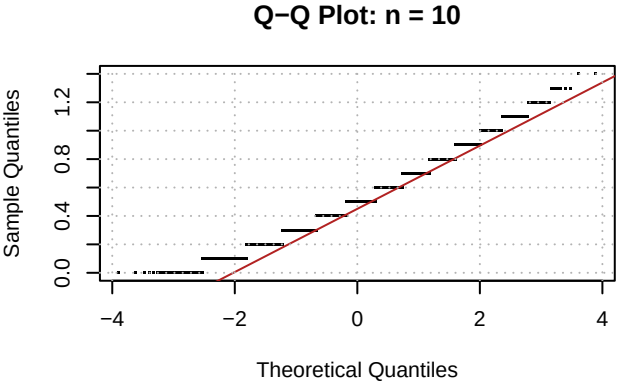

Sampling Distribution of Mean: n = 6
Density
0.0
1.0
2.0
0.0
0.5
1.0
1.5
Sample mean

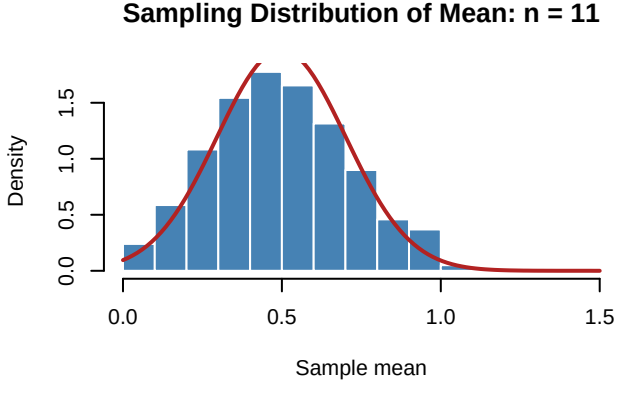

Q−Q Plot: n = 6
Sample Quantiles
0.0
0.5
1.0
1.5
−4
−2
0
2
4
Theoretical Quantiles

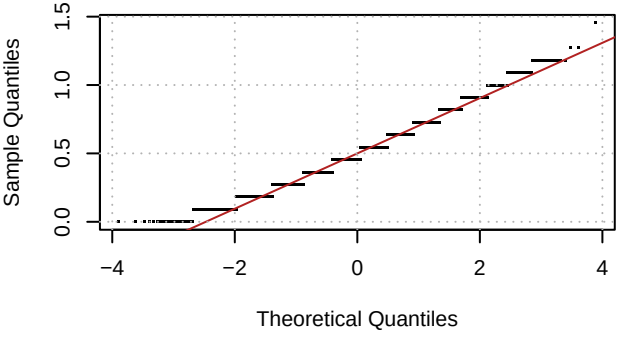

Sampling Distribution of Mean: n = 8
Density
0.0
1.0
2.0
0.0
0.5
1.0
1.5
Sample mean

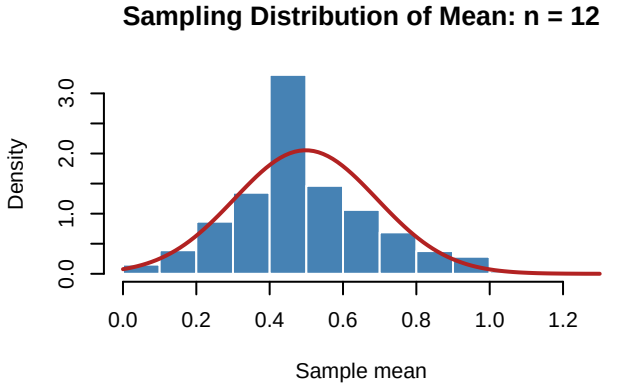

Q−Q Plot: n = 8
Sample Quantiles
0.0
0.5
1.0
1.5
−4
−2
0
2
4
Theoretical Quantiles

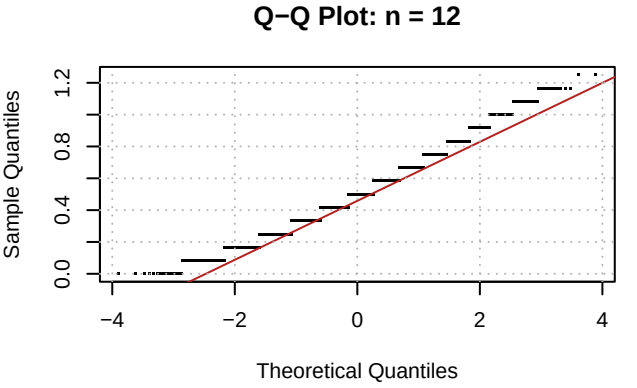

Sampling Distribution of Mean: n = 10
Density
0.0
0.5
1.0
1.5
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Sample mean

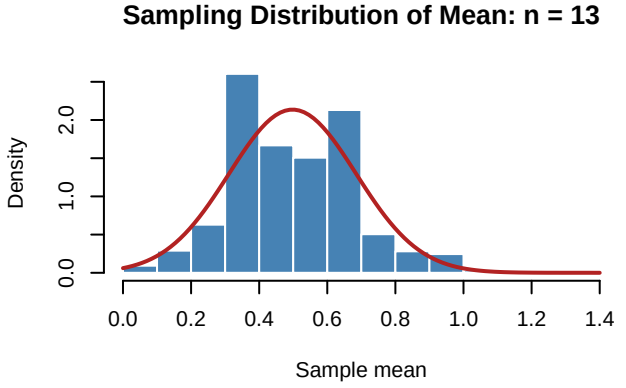

Q−Q Plot: n = 10
Sample Quantiles
0.0
0.4
0.8
1.2
−4
−2
0
2
4
Theoretical Quantiles

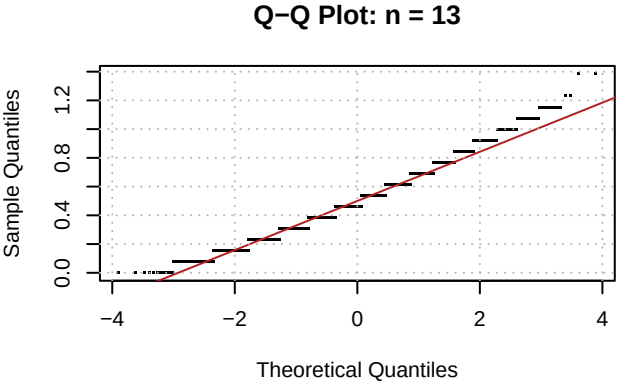

Sampling Distribution of Mean: n = 11
Density
0.0
0.5
1.0
1.5
0.0
0.5
1.0
1.5
Sample mean

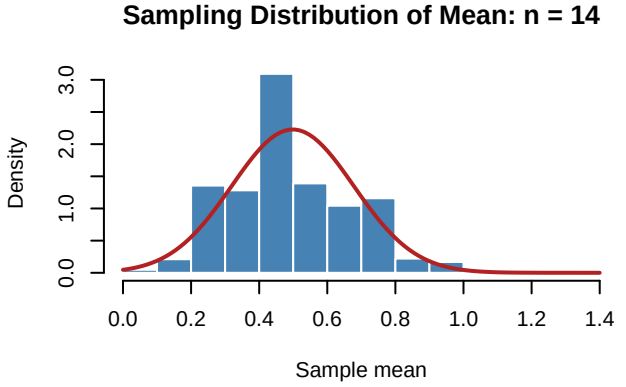

Q−Q Plot: n = 11
Sample Quantiles
0.0
0.5
1.0
1.5
−4
−2
0
2
4
Theoretical Quantiles

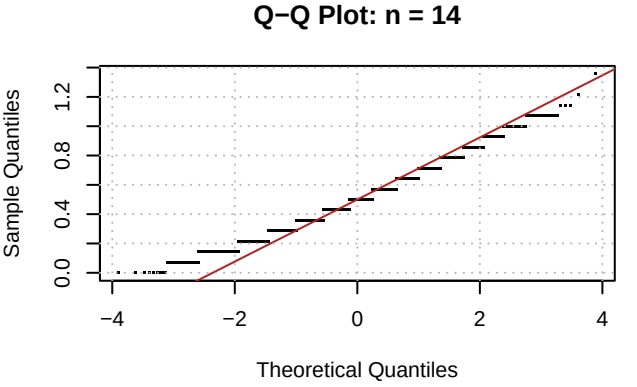

Sampling Distribution of Mean: n = 12
Density
0.0
1.0
2.0
3.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Sample mean

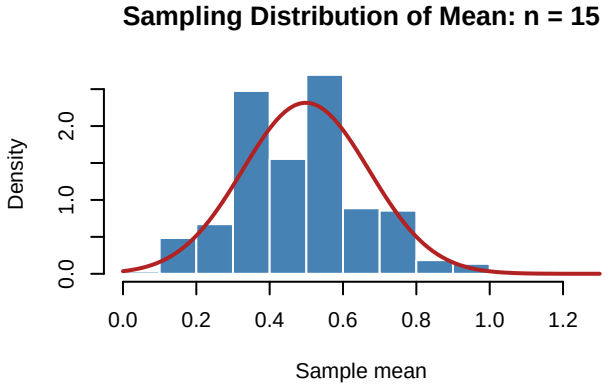

Q−Q Plot: n = 12
Sample Quantiles
0.0
0.4
0.8
1.2
−4
−2
0
2
4
Theoretical Quantiles

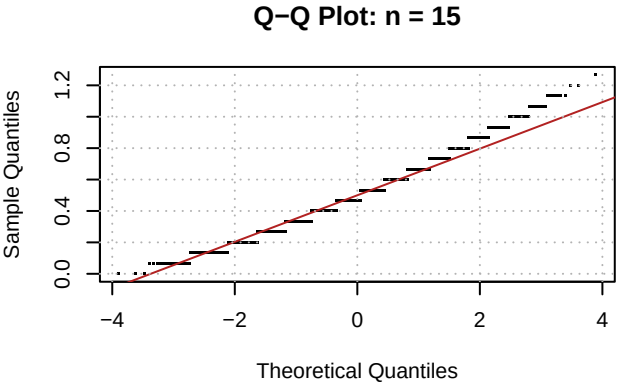

Sampling Distribution of Mean: n = 13
Density
0.0
1.0
2.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Sample mean

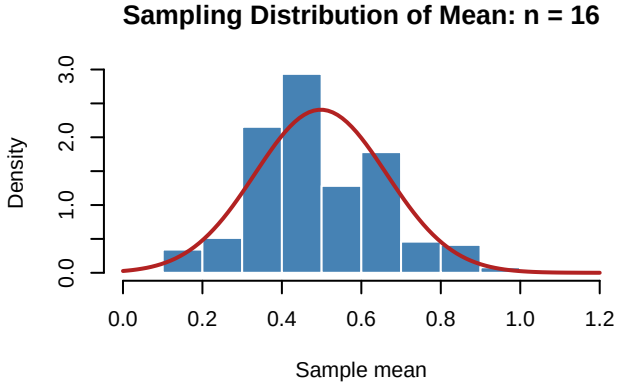

Q−Q Plot: n = 13
Sample Quantiles
0.0
0.4
0.8
1.2
−4
−2
0
2
4
Theoretical Quantiles

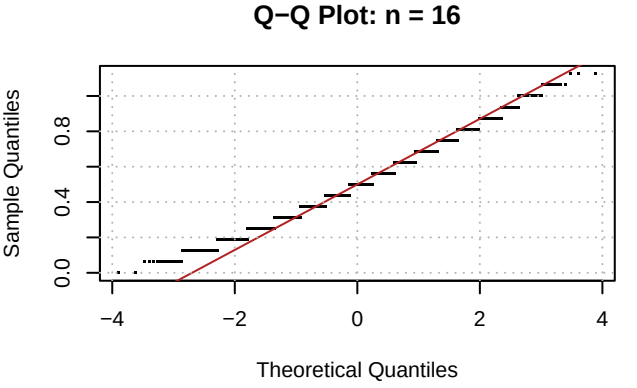

Sampling Distribution of Mean: n = 14
Density
0.0
1.0
2.0
3.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
Sample mean

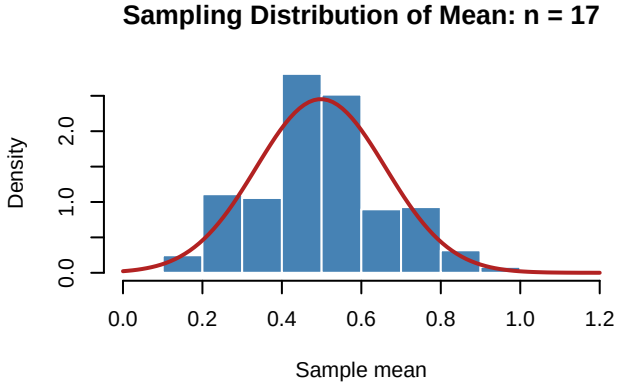

Q−Q Plot: n = 14
Sample Quantiles
0.0
0.4
0.8
1.2
−4
−2
0
2
4
Theoretical Quantiles

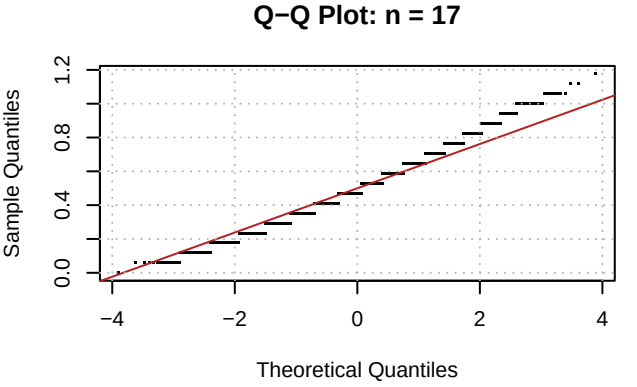

Sampling Distribution of Mean: n = 15
Density
0.0
1.0
2.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Sample mean

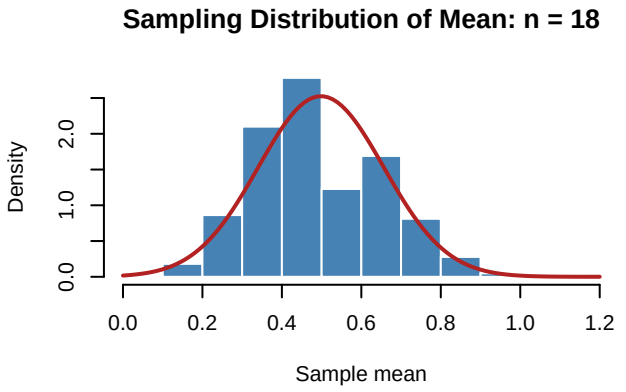

Q−Q Plot: n = 15
Sample Quantiles
0.0
0.4
0.8
1.2
−4
−2
0
2
4
Theoretical Quantiles

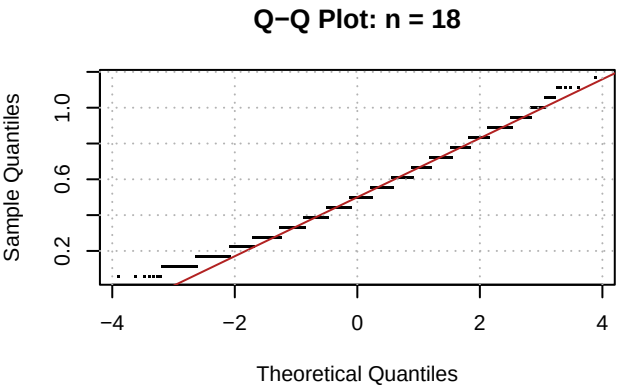

Sampling Distribution of Mean: n = 16
Density
0.0
1.0
2.0
3.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Sample mean

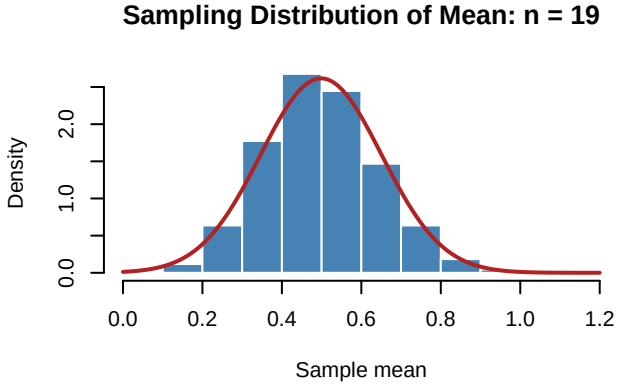

Q−Q Plot: n = 16
Sample Quantiles
0.0
0.4
0.8
−4
−2
0
2
4
Theoretical Quantiles

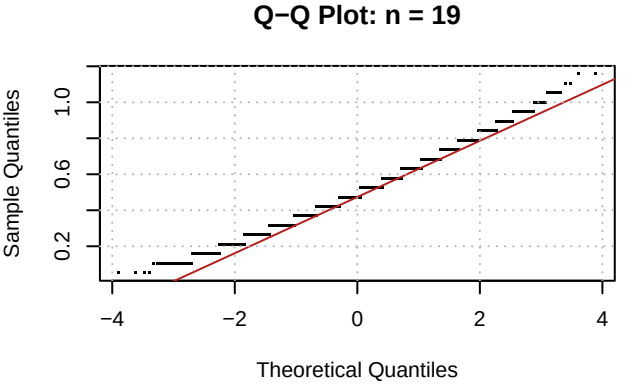

Sampling Distribution of Mean: n = 17
Density
0.0
1.0
2.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Sample mean

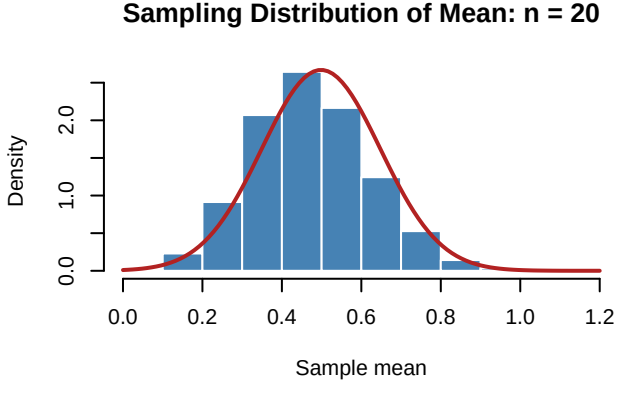

Q−Q Plot: n = 17
Sample Quantiles
0.0
0.4
0.8
1.2
−4
−2
0
2
4
Theoretical Quantiles

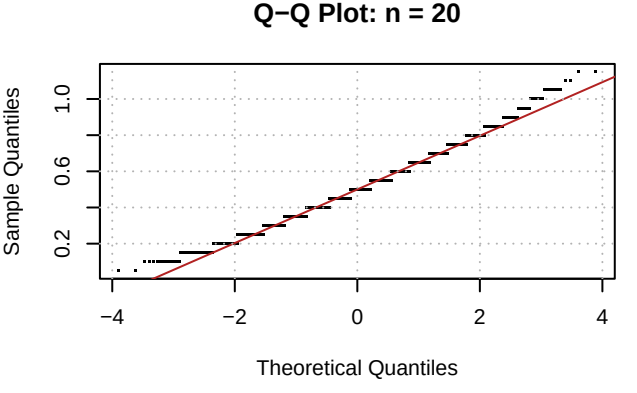

Sampling Distribution of Mean: n = 18
Density
0.0
1.0
2.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Sample mean

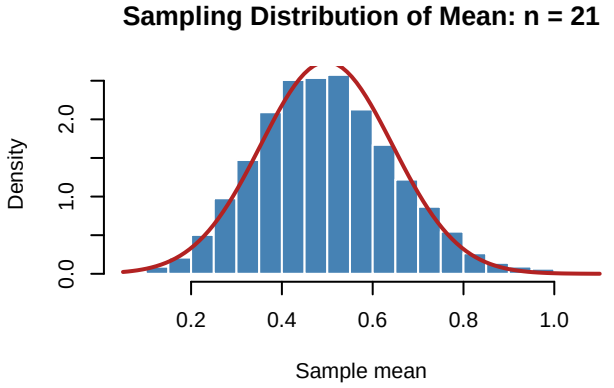

Q−Q Plot: n = 18
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

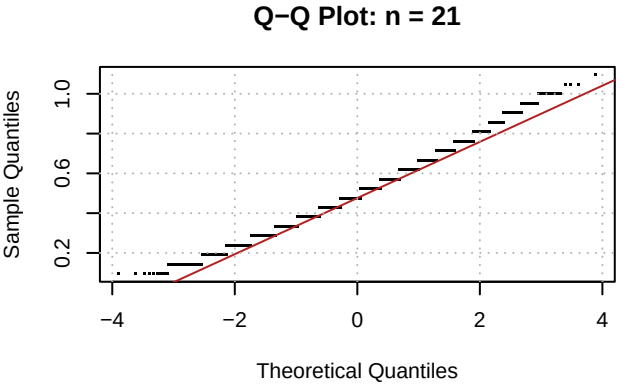

Sampling Distribution of Mean: n = 19
Density
0.0
1.0
2.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Sample mean

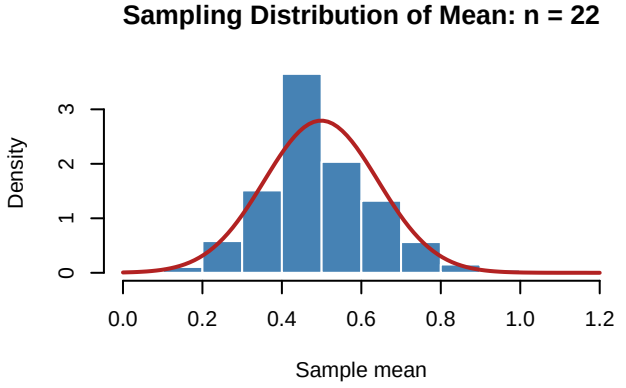

Q−Q Plot: n = 19
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

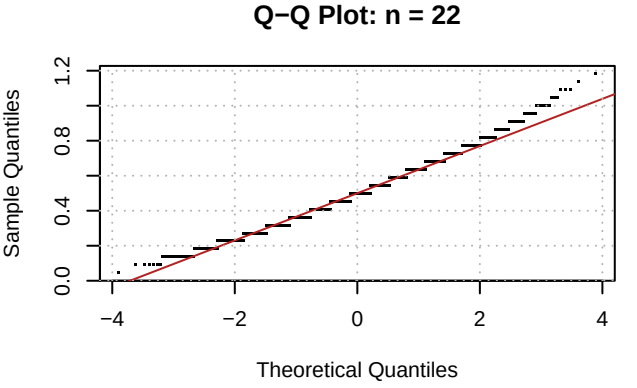

Sampling Distribution of Mean: n = 20
Density
0.0
1.0
2.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Sample mean

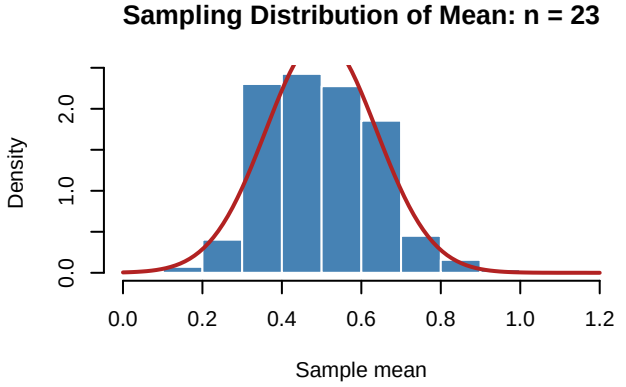

Q−Q Plot: n = 20
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

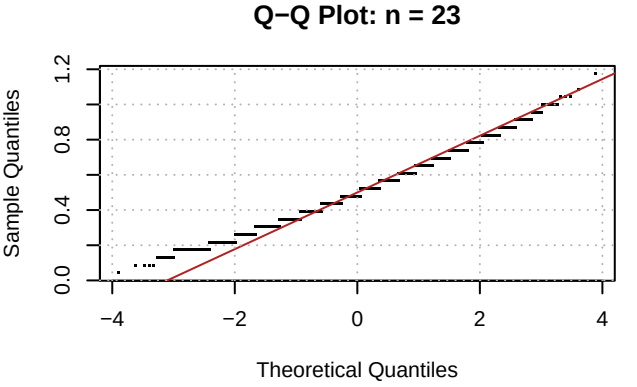

Sampling Distribution of Mean: n = 21
Density
0.0
1.0
2.0
0.2
0.4
0.6
0.8
1.0
Sample mean

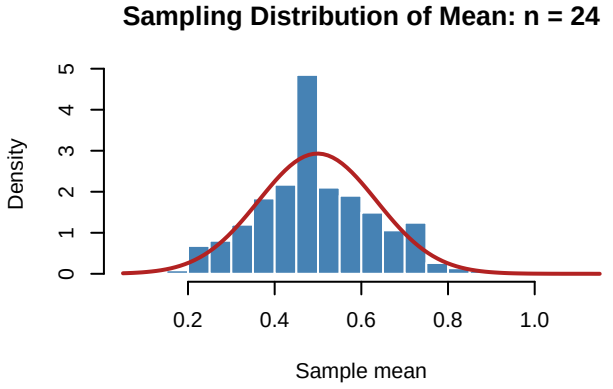

Q−Q Plot: n = 21
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

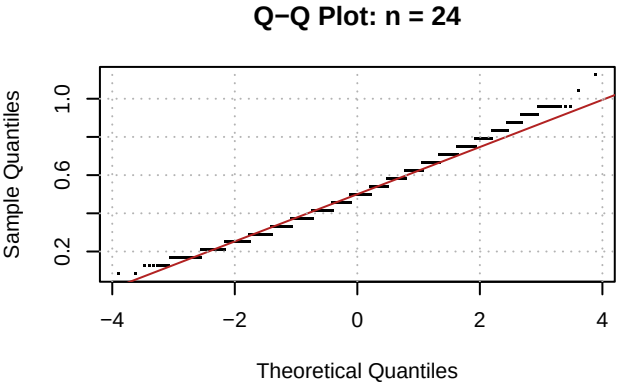

Sampling Distribution of Mean: n = 22
Density
0
1
2
3
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Sample mean

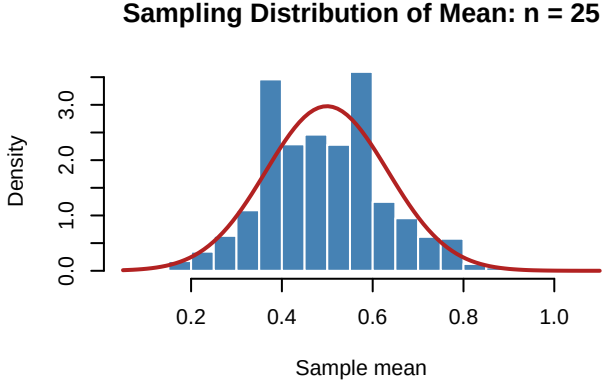

Q−Q Plot: n = 22
Sample Quantiles
0.0
0.4
0.8
1.2
−4
−2
0
2
4
Theoretical Quantiles

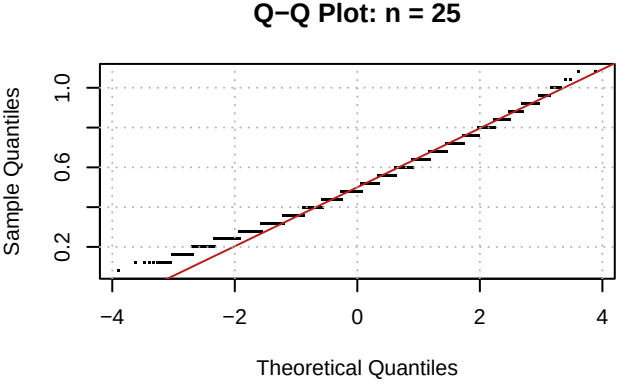

Sampling Distribution of Mean: n = 23
Density
0.0
1.0
2.0
0.0
0.2
0.4
0.6
0.8
1.0
1.2
Sample mean

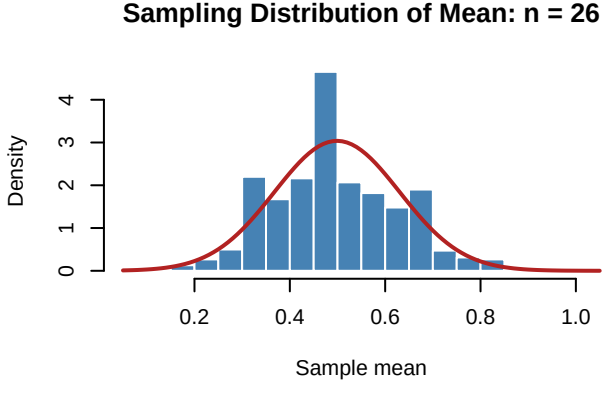

Q−Q Plot: n = 23
Sample Quantiles
0.0
0.4
0.8
1.2
−4
−2
0
2
4
Theoretical Quantiles

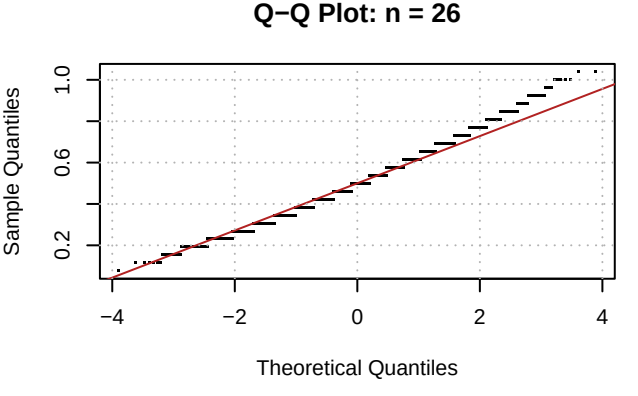

Sampling Distribution of Mean: n = 24
Density
0
1
2
3
4
5
0.2
0.4
0.6
0.8
1.0
Sample mean

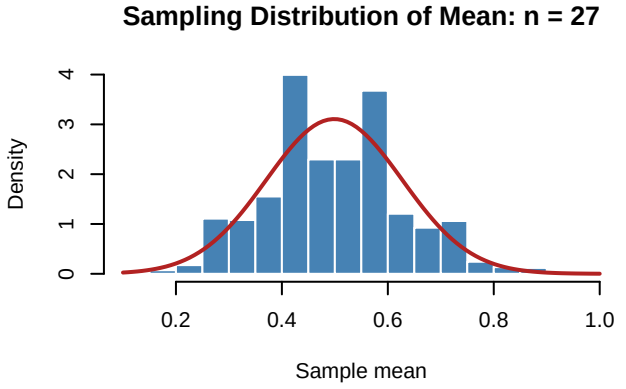

Q−Q Plot: n = 24
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

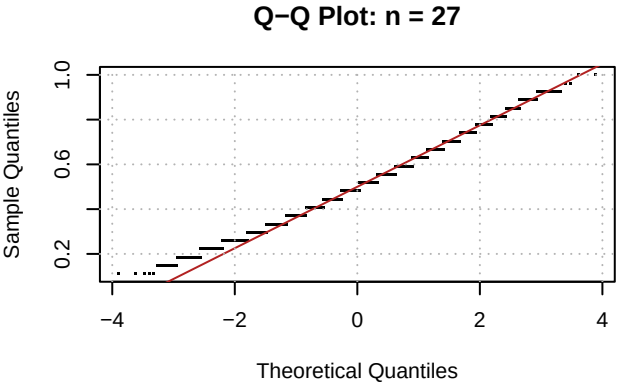

Sampling Distribution of Mean: n = 25
Density
0.0
1.0
2.0
3.0
0.2
0.4
0.6
0.8
1.0
Sample mean

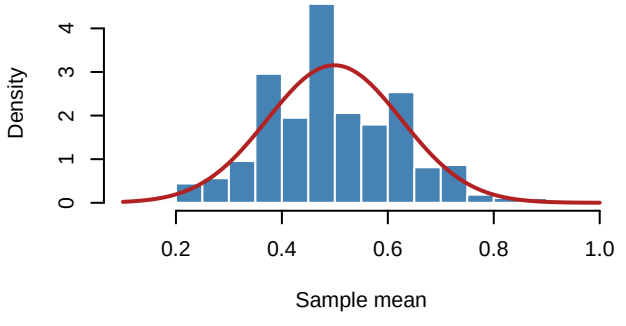

Q−Q Plot: n = 25
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

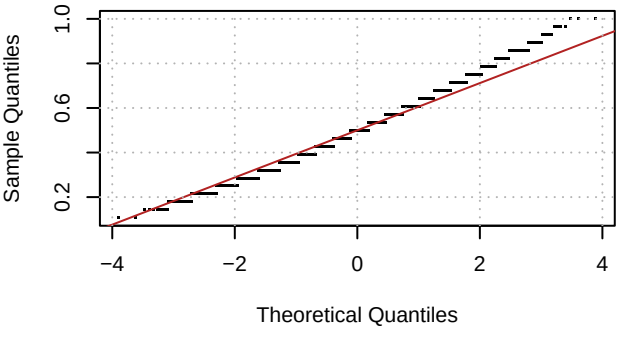

Sampling Distribution of Mean: n = 26
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
Sample mean

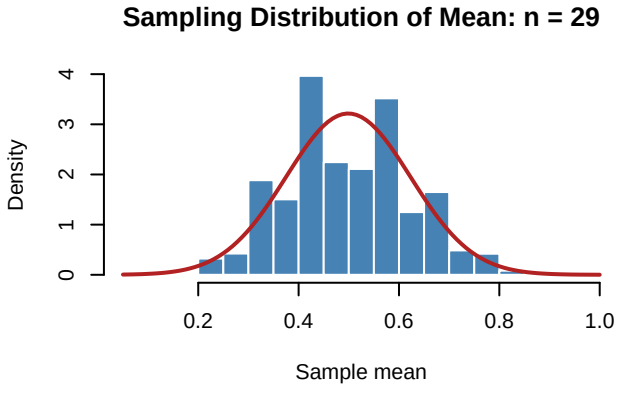

Q−Q Plot: n = 26
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

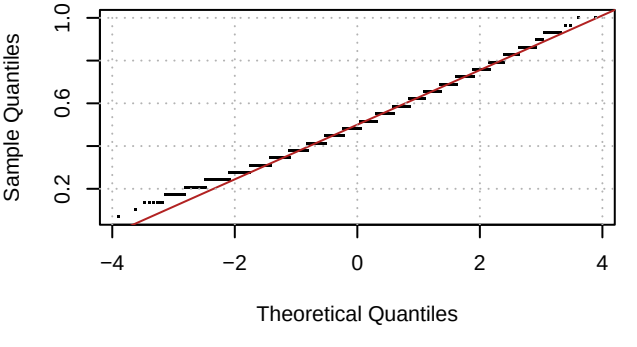

Sampling Distribution of Mean: n = 27
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
Sample mean

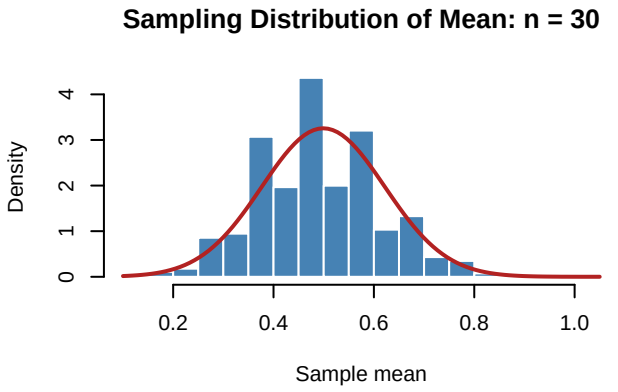

Q−Q Plot: n = 27
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

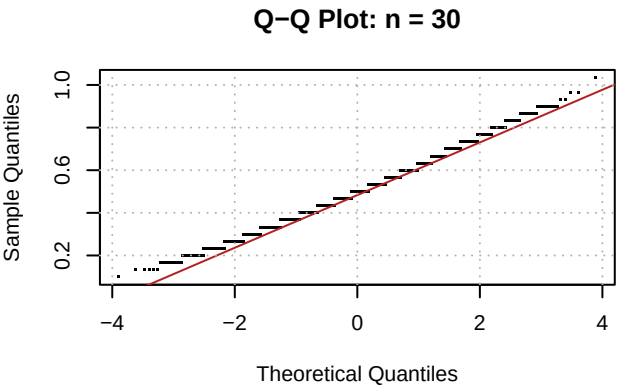

Sampling Distribution of Mean: n = 28
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
Sample mean

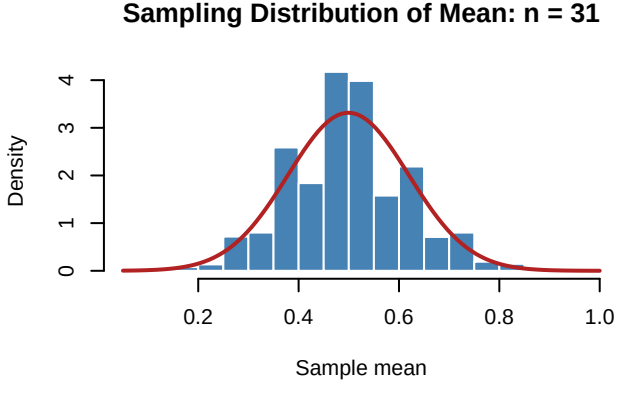

Q−Q Plot: n = 28
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

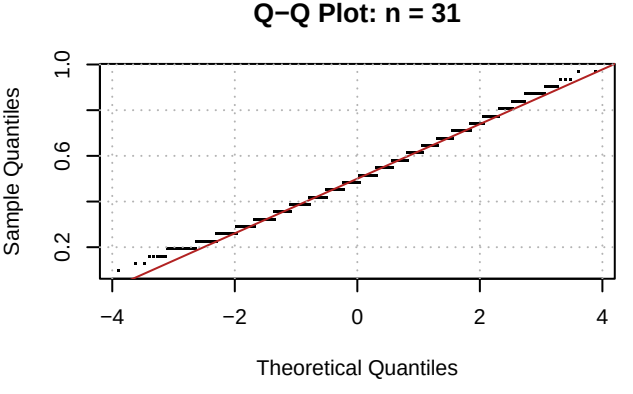

Sampling Distribution of Mean: n = 29
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
Sample mean

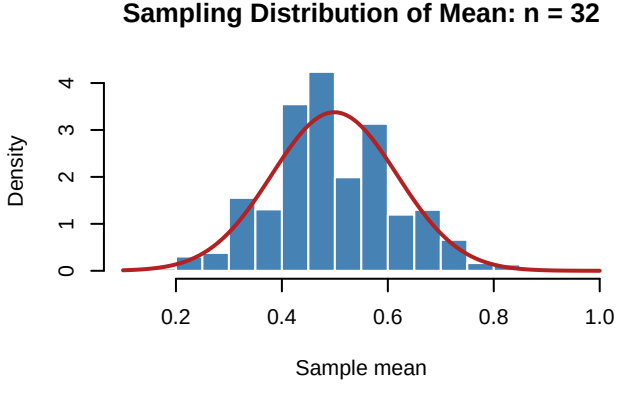

Q−Q Plot: n = 29
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

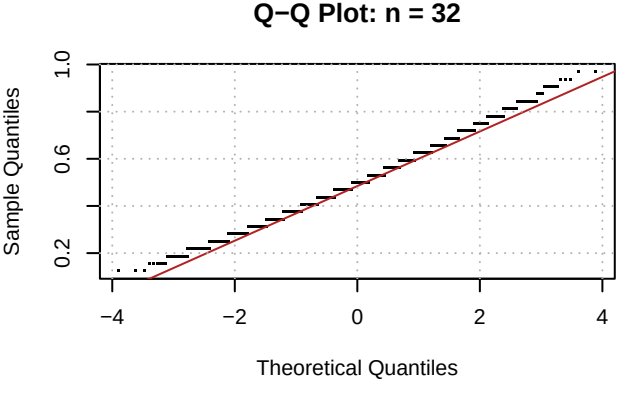

Sampling Distribution of Mean: n = 30
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
Sample mean

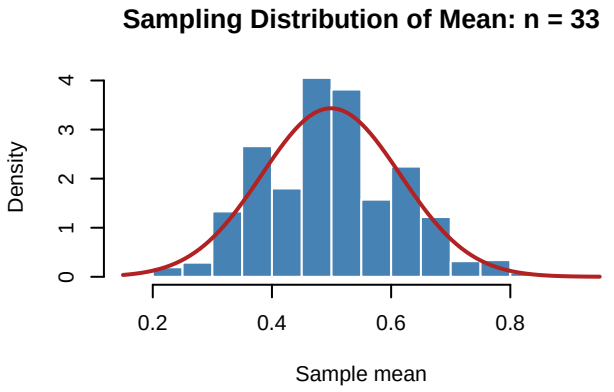

Q−Q Plot: n = 30
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

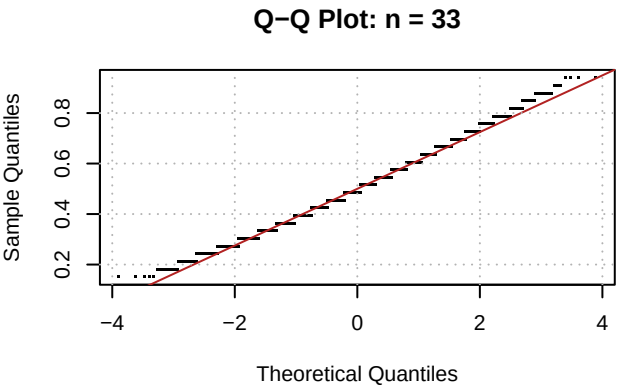

Sampling Distribution of Mean: n = 31
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
Sample mean

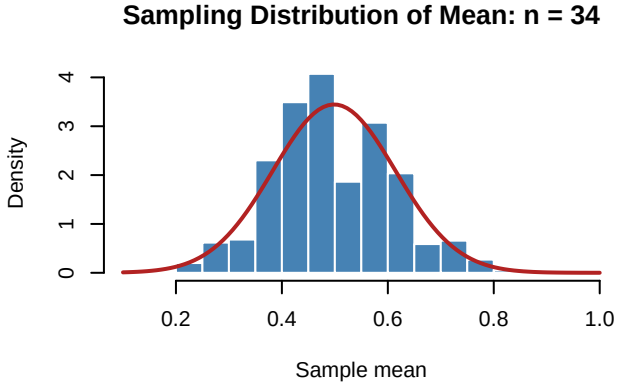

Q−Q Plot: n = 31
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

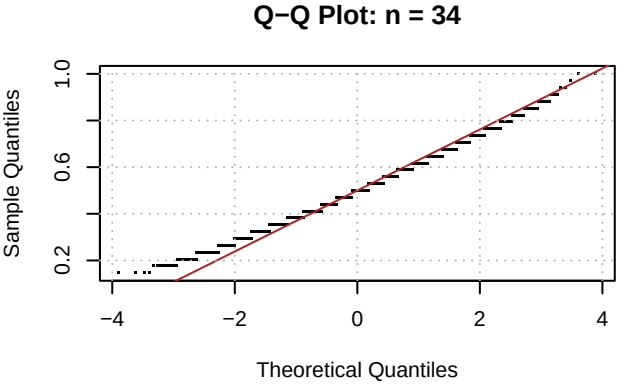

Sampling Distribution of Mean: n = 32
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
Sample mean

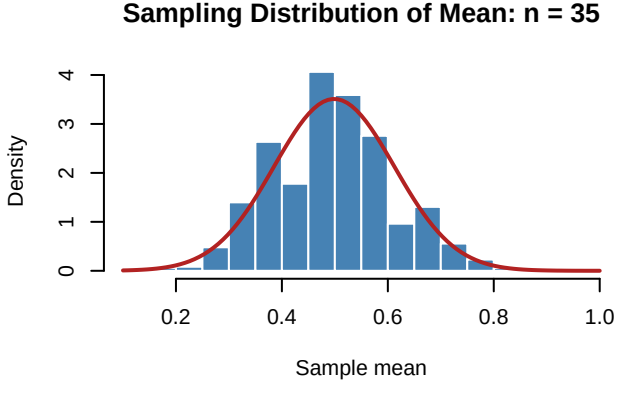

Q−Q Plot: n = 32
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

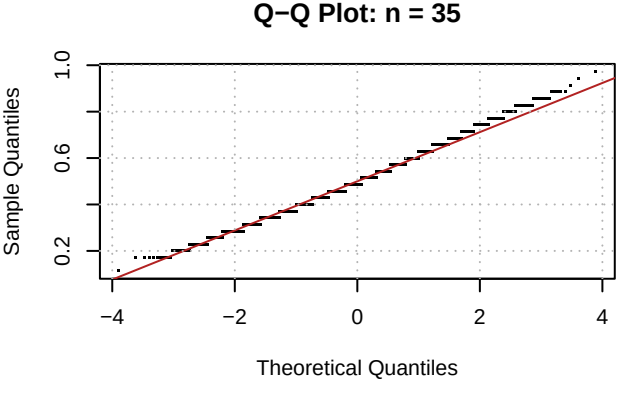

Sampling Distribution of Mean: n = 33
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
Sample mean

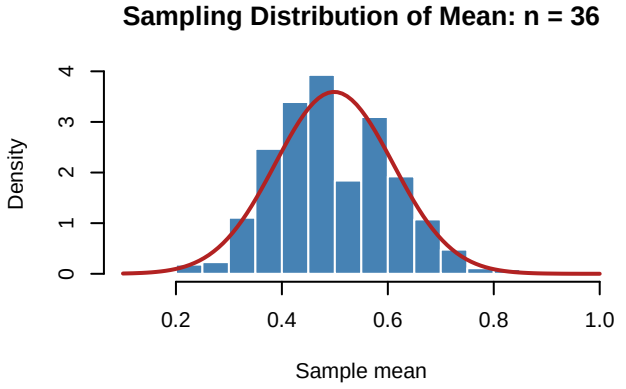

Q−Q Plot: n = 33
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

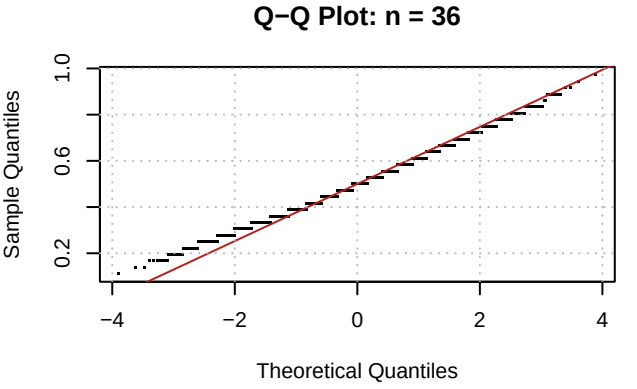

Sampling Distribution of Mean: n = 34
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
Sample mean

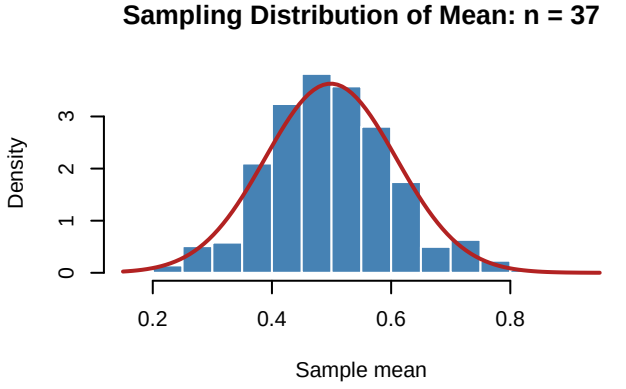

Q−Q Plot: n = 34
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

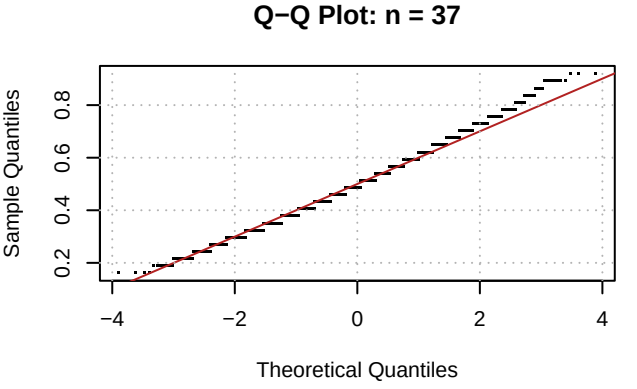

Sampling Distribution of Mean: n = 35
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
Sample mean

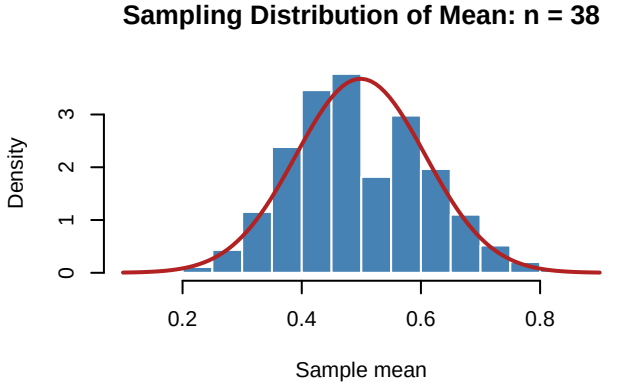

Q−Q Plot: n = 35
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

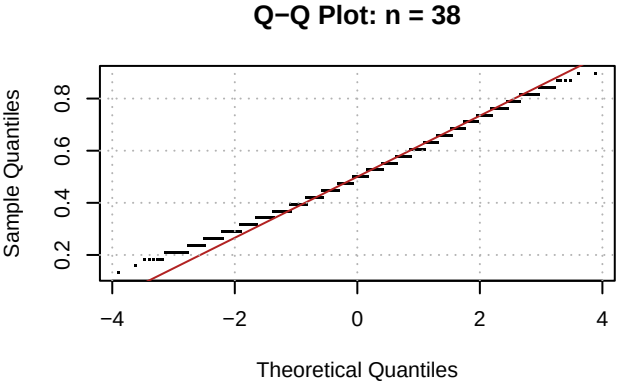

Sampling Distribution of Mean: n = 36
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
1.0
Sample mean

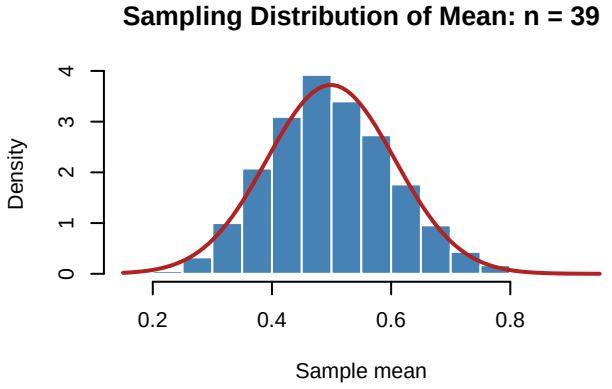

Q−Q Plot: n = 36
Sample Quantiles
0.2
0.6
1.0
−4
−2
0
2
4
Theoretical Quantiles

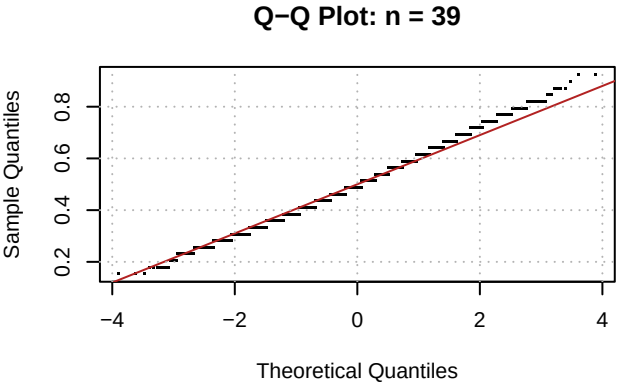

Sampling Distribution of Mean: n = 37
Density
0
1
2
3
0.2
0.4
0.6
0.8
Sample mean

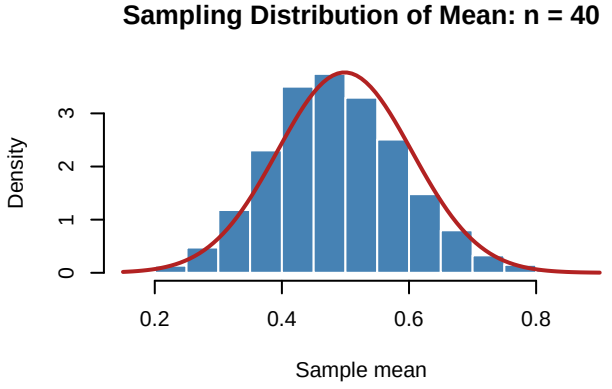

Q−Q Plot: n = 37
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

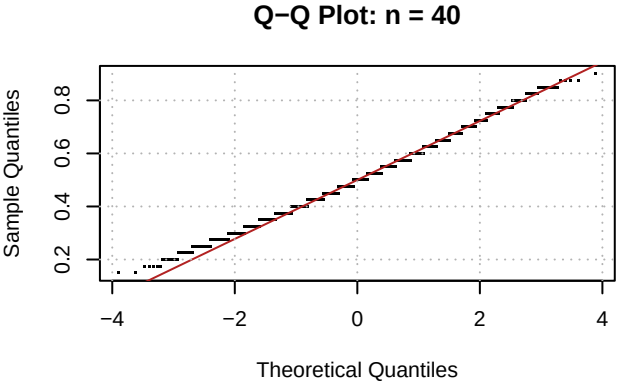

Sampling Distribution of Mean: n = 38
Density
0
1
2
3
0.2
0.4
0.6
0.8
Sample mean

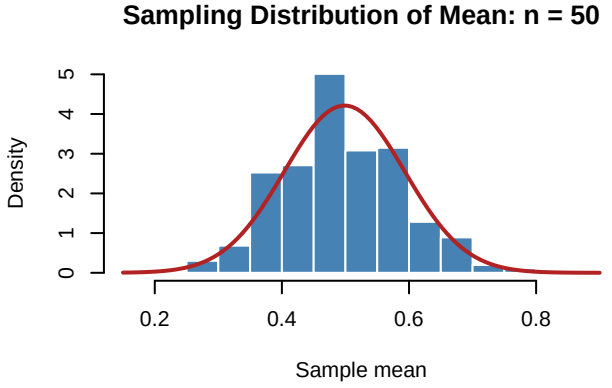

Q−Q Plot: n = 38
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

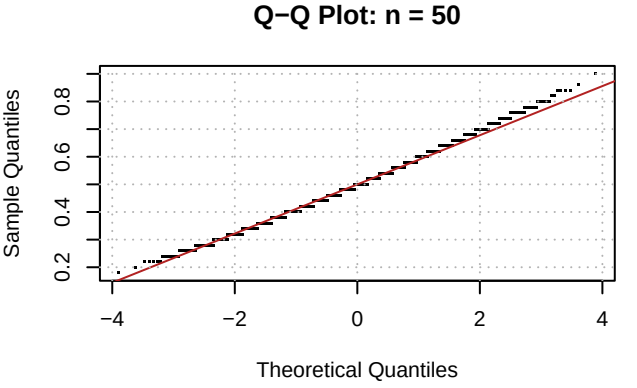

Sampling Distribution of Mean: n = 39
Density
0
1
2
3
4
0.2
0.4
0.6
0.8
Sample mean

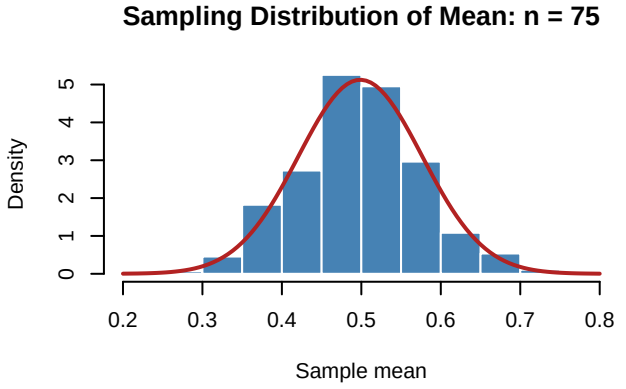

Q−Q Plot: n = 39
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

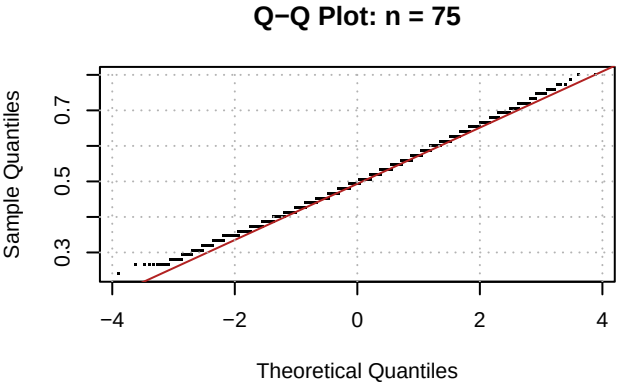

Sampling Distribution of Mean: n = 40
Density
0
1
2
3
0.2
0.4
0.6
0.8
Sample mean

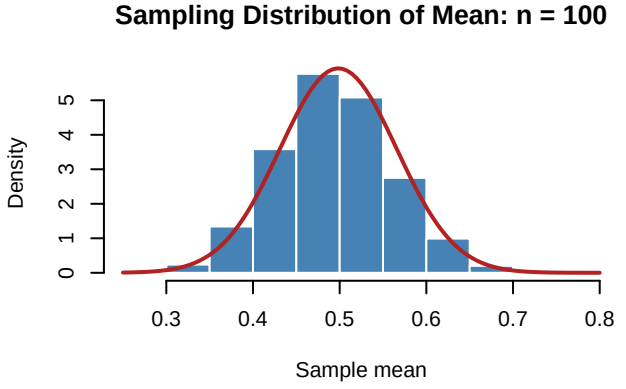

Q−Q Plot: n = 40
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

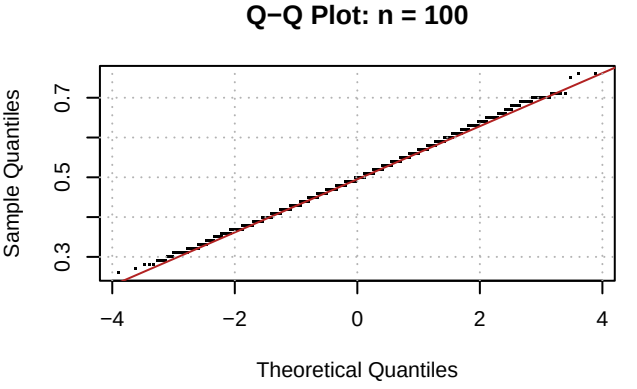

Sampling Distribution of Mean: n = 50
Density
0
1
2
3
4
5
0.2
0.4
0.6
0.8
Sample mean

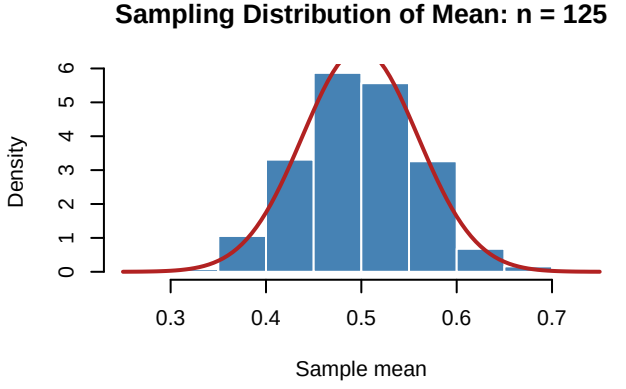

Q−Q Plot: n = 50
Sample Quantiles
0.2
0.4
0.6
0.8
−4
−2
0
2
4
Theoretical Quantiles

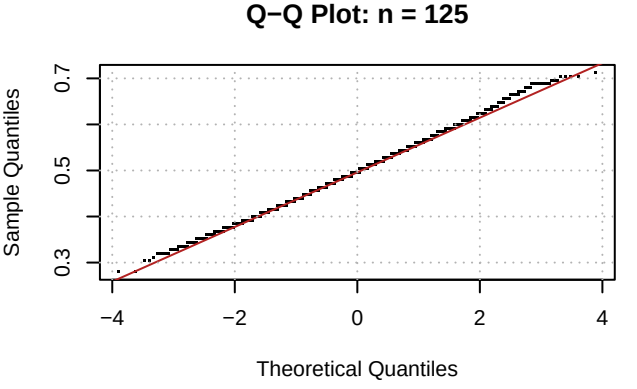

Sampling Distribution of Mean: n = 75
Density
0 1 2 3 4 5
0.2 0.3 0.4 0.5 0.6 0.7 0.8
Sample mean

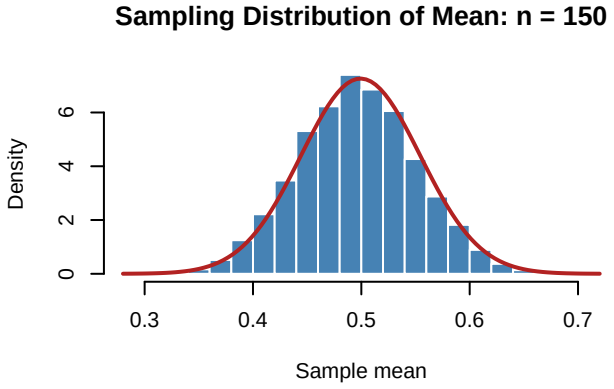

Q−Q Plot: n = 75
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

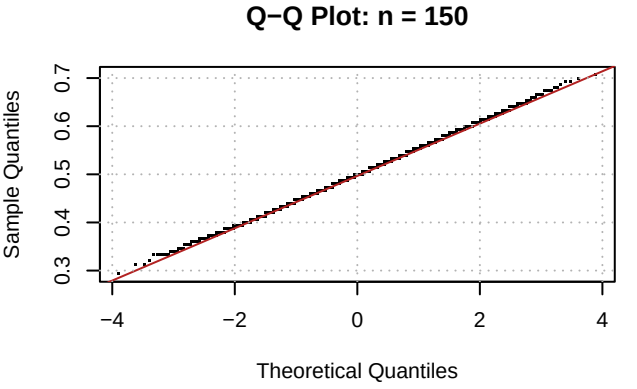

Sampling Distribution of Mean: n = 100
Density
0 1 2 3 4 5
0.3 0.4 0.5 0.6 0.7 0.8
Sample mean

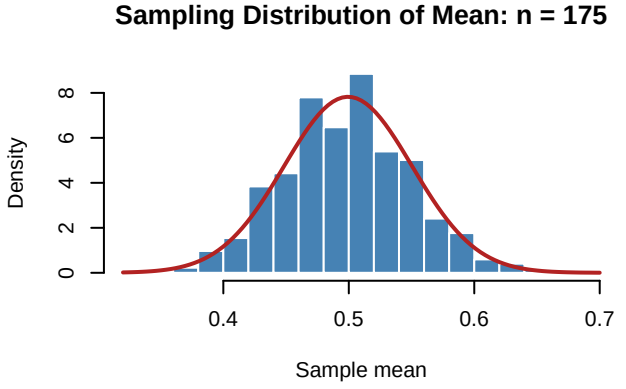

Q−Q Plot: n = 100
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

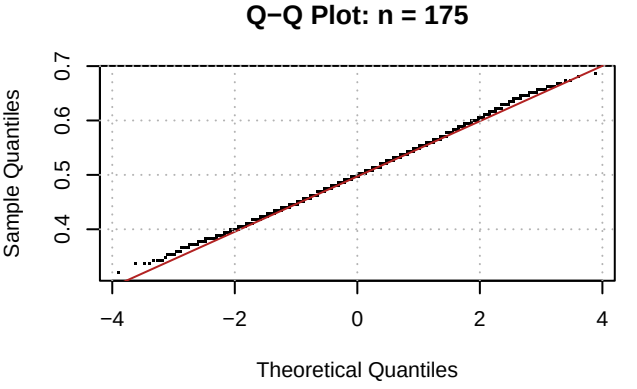

Sampling Distribution of Mean: n = 125
Density
0 1 2 3 4 5 6
0.3 0.4 0.5 0.6 0.7
Sample mean

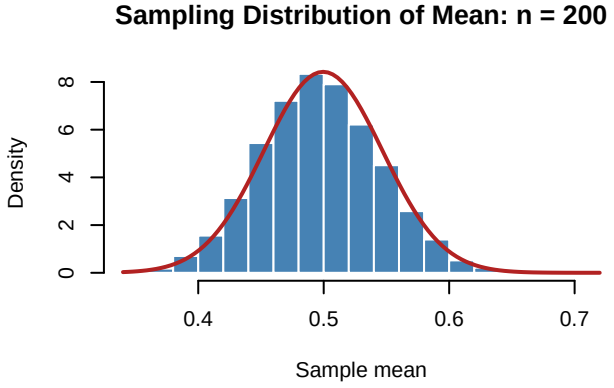

Q−Q Plot: n = 125
Sample Quantiles
0.3 0.5 0.7
−4 −2 0 2 4
Theoretical Quantiles

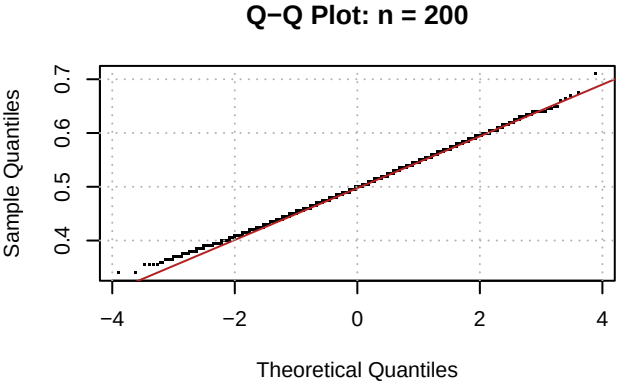

Sampling Distribution of Mean: n = 150
Density
0
2
4
6
0.3
0.4
0.5
0.6
0.7
Sample mean

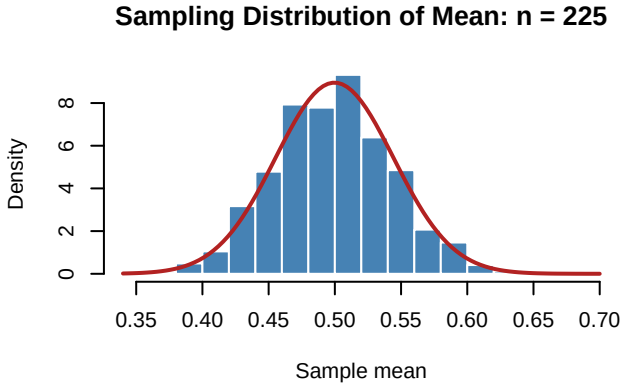

Q−Q Plot: n = 150
Sample Quantiles
0.3
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

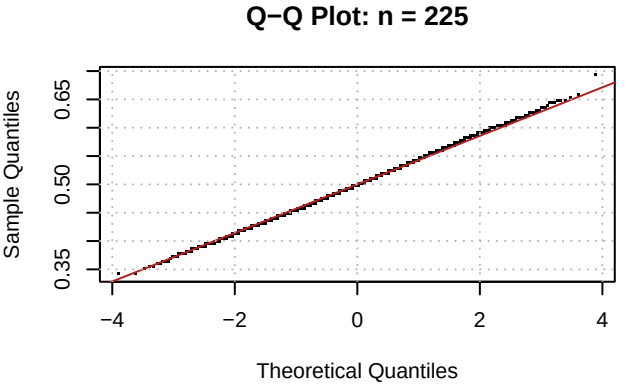

Sampling Distribution of Mean: n = 175
Density
0
2
4
6
8
0.4
0.5
0.6
0.7
Sample mean

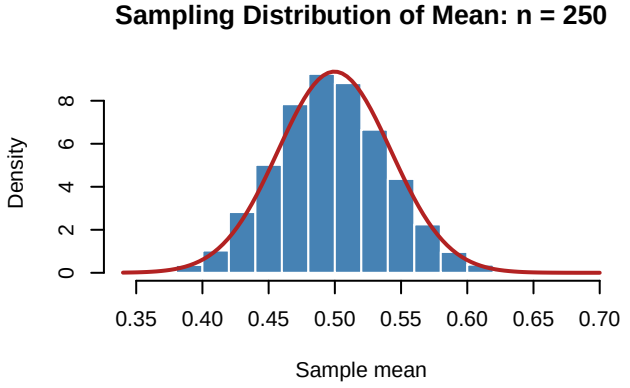

Q−Q Plot: n = 175
Sample Quantiles
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

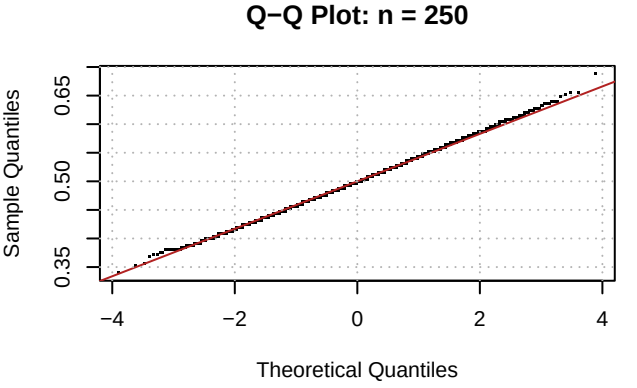

Sampling Distribution of Mean: n = 200
Density
0
2
4
6
8
0.4
0.5
0.6
0.7
Sample mean

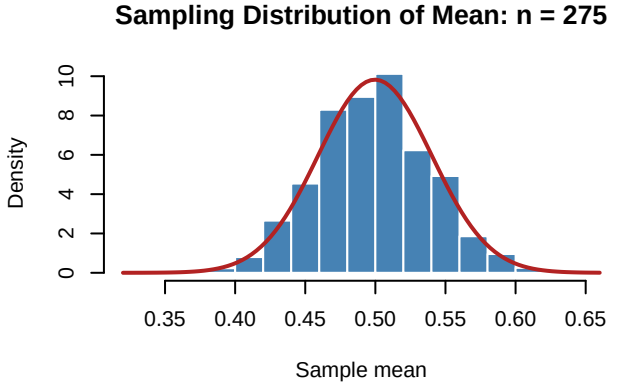

Q−Q Plot: n = 200
Sample Quantiles
0.4
0.5
0.6
0.7
−4
−2
0
2
4
Theoretical Quantiles

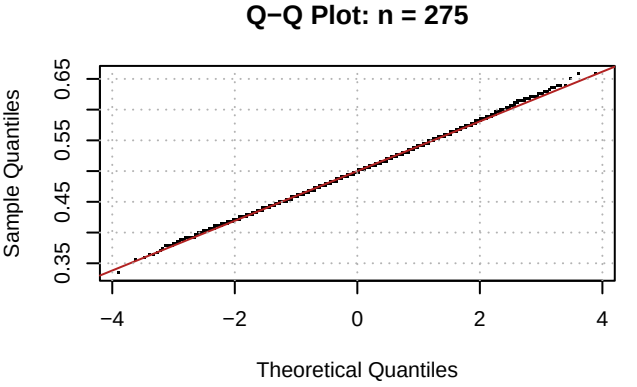

Sampling Distribution of Mean: n = 225
Density
0
2
4
6
8
0.35
0.40
0.45
0.50
0.55
0.60
0.65
0.70
Sample mean

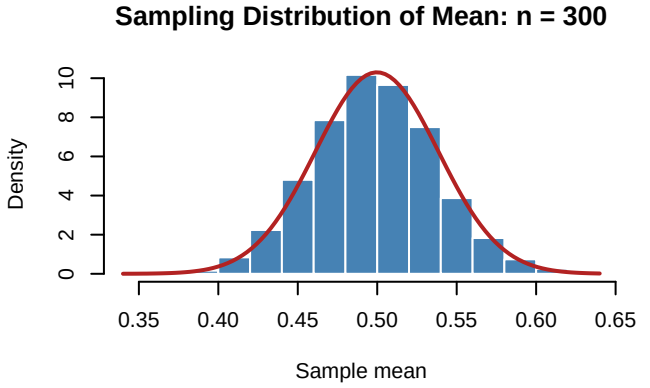

Q−Q Plot: n = 225
Sample Quantiles
0.35
0.50
0.65
−4
−2
0
2
4
Theoretical Quantiles

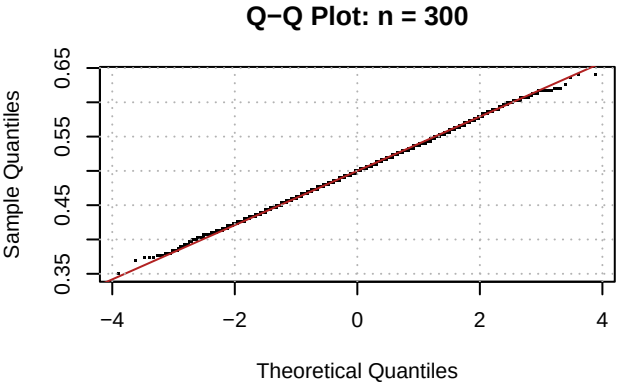

Sampling Distribution of Mean: n = 250
Density
0
2
4
6
8
0.35
0.40
0.45
0.50
0.55
0.60
0.65
0.70
Sample mean

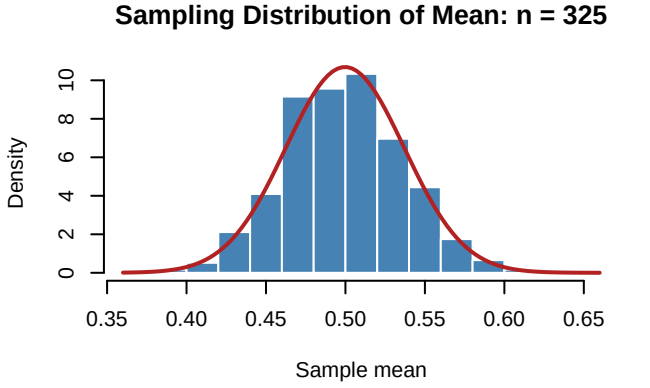

Q−Q Plot: n = 250
Sample Quantiles
0.35
0.50
0.65
−4
−2
0
2
4
Theoretical Quantiles

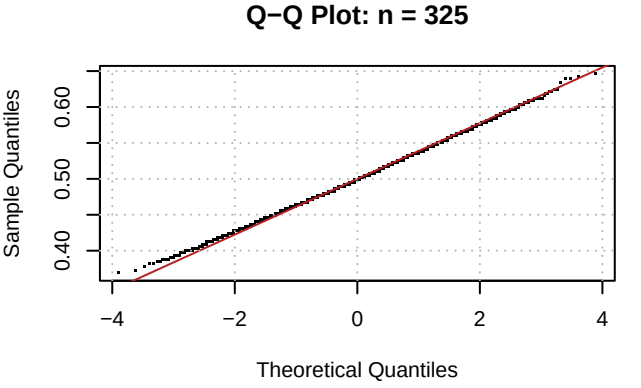

Sampling Distribution of Mean: n = 275
Density
0
2
4
6
8
10
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

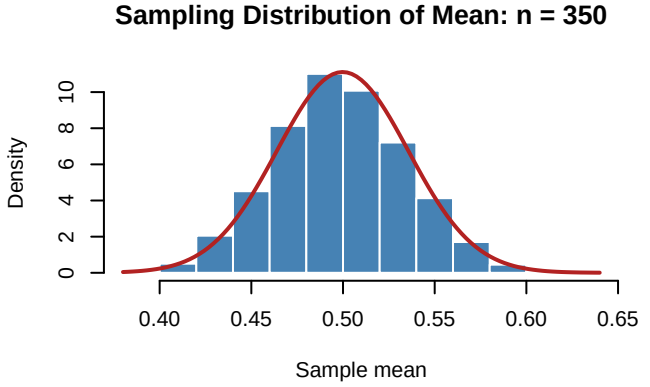

Q−Q Plot: n = 275
Sample Quantiles
0.35
0.45
0.55
0.65
−4
−2
0
2
4
Theoretical Quantiles

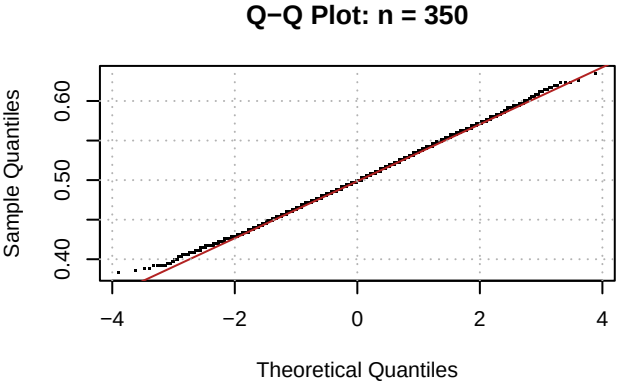

Sampling Distribution of Mean: n = 300
Density
0
2
4
6
8
10
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

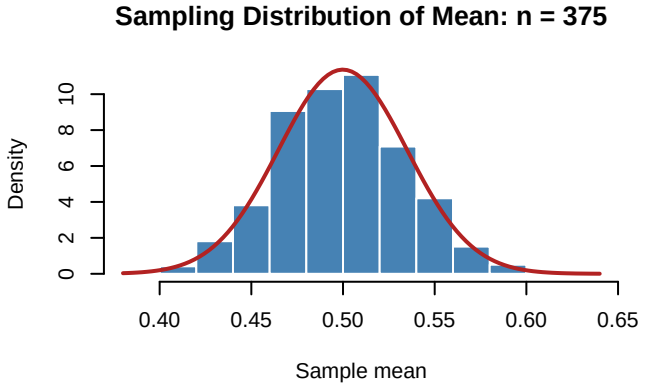

Q−Q Plot: n = 300
Sample Quantiles
0.35
0.45
0.55
0.65
−4
−2
0
2
4
Theoretical Quantiles

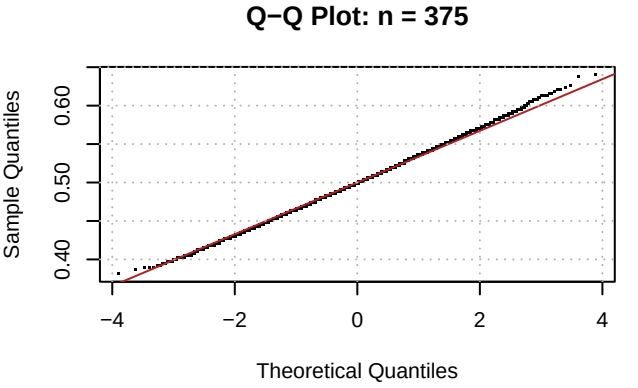

Sampling Distribution of Mean: n = 325
Density
0
2
4
6
8
10
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

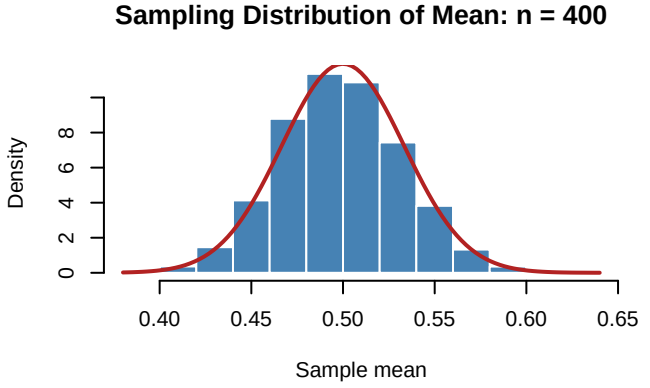

Q−Q Plot: n = 325
Sample Quantiles
0.40
0.50
0.60
−4
−2
0
2
4
Theoretical Quantiles

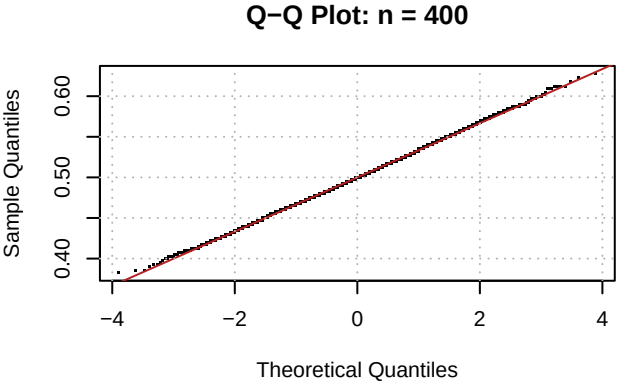

Sampling Distribution of Mean: n = 350
Density
0
2
4
6
8
10
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

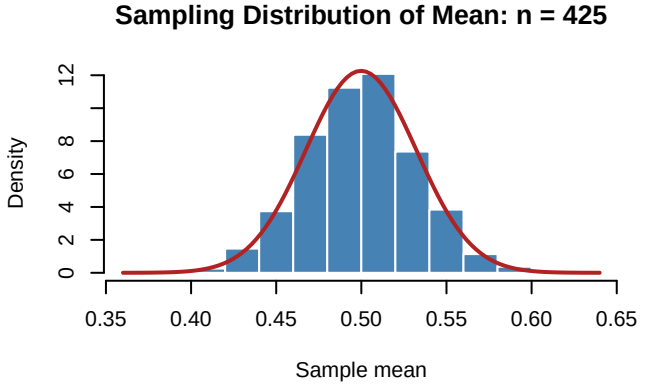

Q−Q Plot: n = 350
Sample Quantiles
0.40
0.50
0.60
−4
−2
0
2
4
Theoretical Quantiles

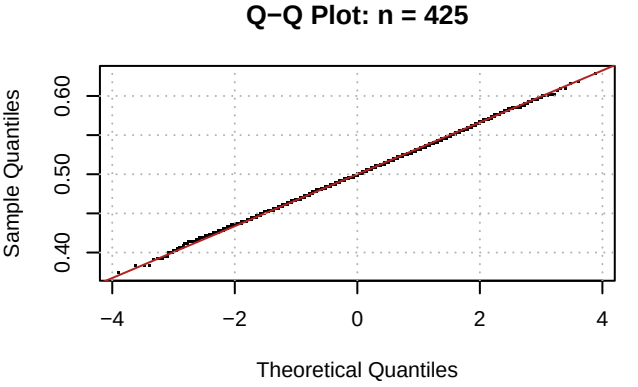

Sampling Distribution of Mean: n = 375
Density
0
2
4
6
8
10
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

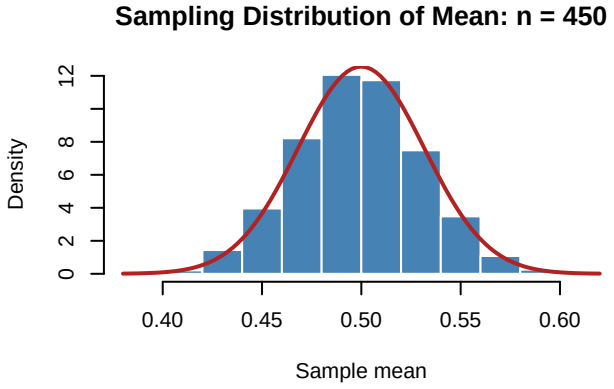

Q−Q Plot: n = 375
Sample Quantiles
0.40
0.50
0.60
−4
−2
0
2
4
Theoretical Quantiles

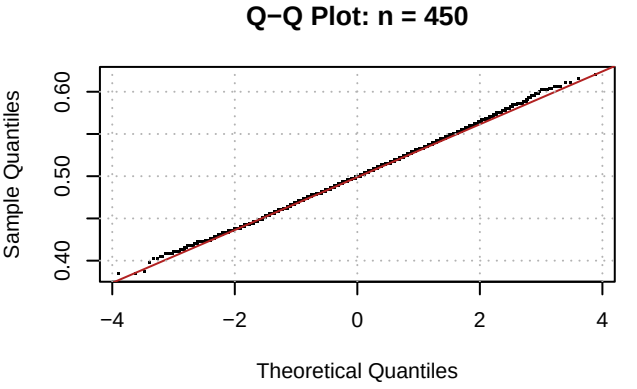

Sampling Distribution of Mean: n = 400
Density
0
2
4
6
8
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

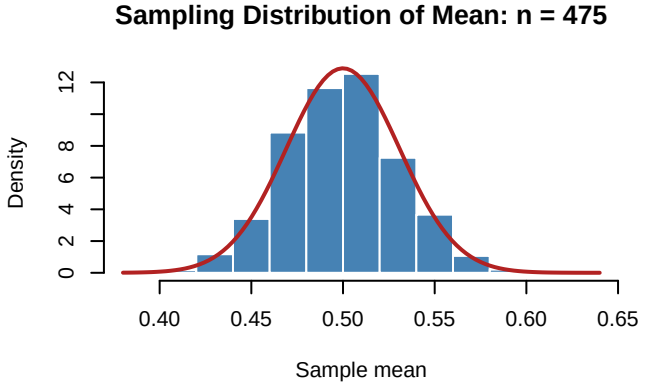

Q−Q Plot: n = 400
Sample Quantiles
0.40
0.50
0.60
−4
−2
0
2
4
Theoretical Quantiles

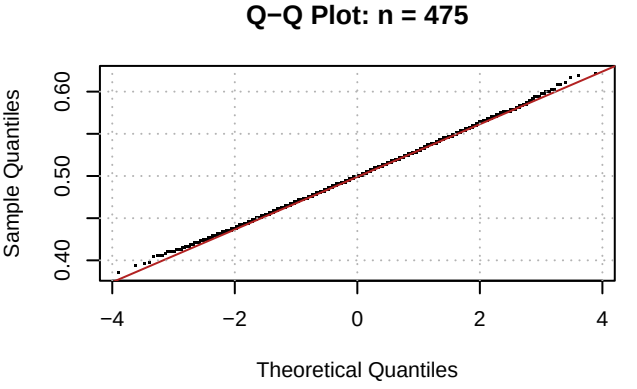

Sampling Distribution of Mean: n = 425
Density
0
2
4
6
8
12
0.35
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

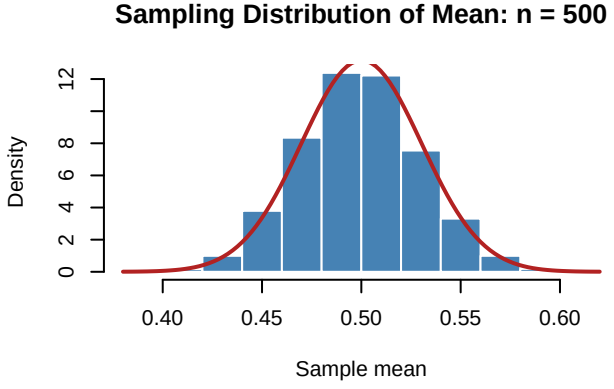

Q−Q Plot: n = 425
Sample Quantiles
0.40
0.50
0.60
−4
−2
0
2
4
Theoretical Quantiles

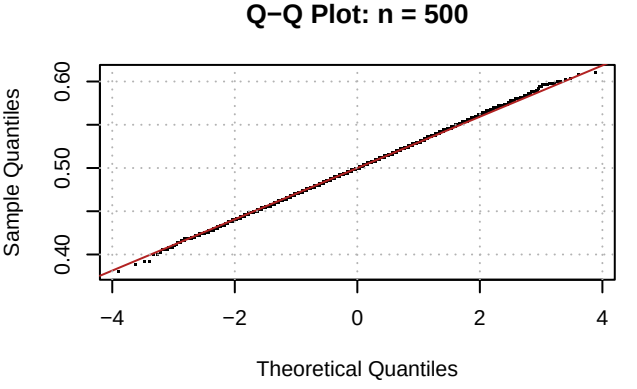

Sampling Distribution of Mean: n = 450
Density
0
2
4
6
8
12
0.40
0.45
0.50
0.55
0.60
Sample mean

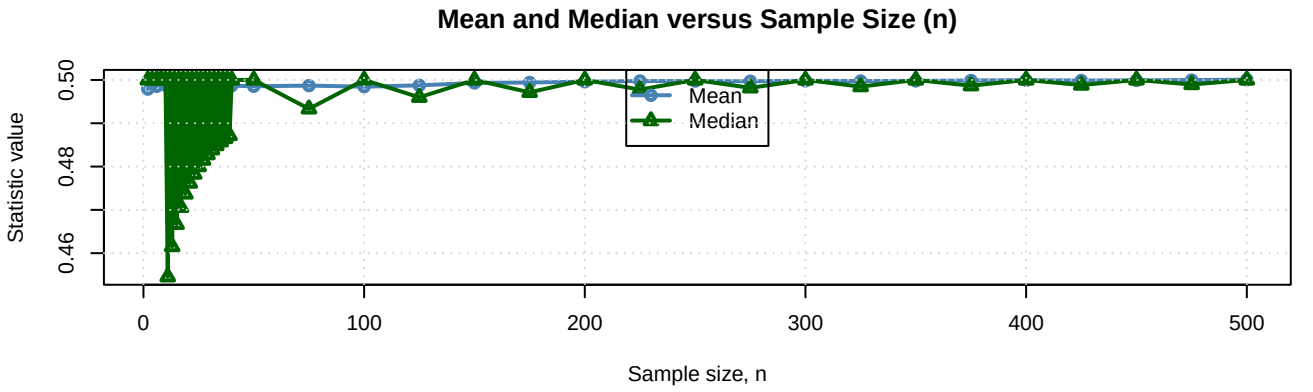

Q−Q Plot: n = 450
Sample Quantiles
0.40
0.50
0.60
−4
−2
0
2
4
Theoretical Quantiles

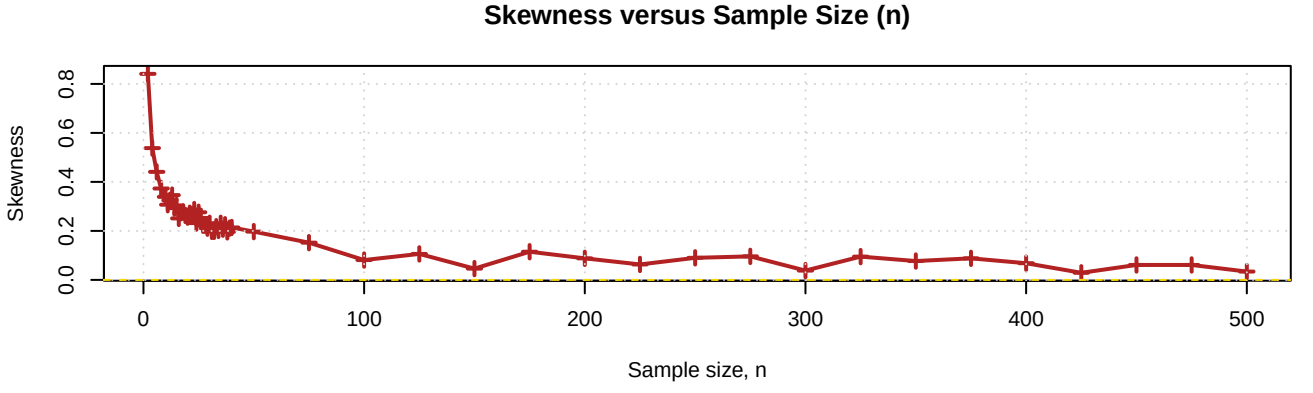

Sampling Distribution of Mean: n = 475
Density
0
2
4
6
8
12
0.40
0.45
0.50
0.55
0.60
0.65
Sample mean

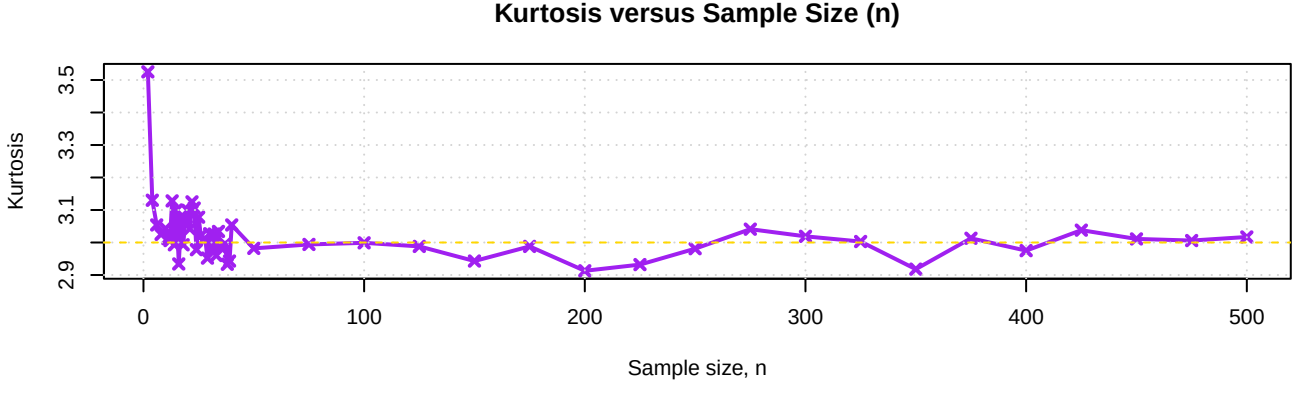

Q−Q Plot: n = 475
Sample Quantiles
0.40
0.50
0.60
−4
−2
0
2
4
Theoretical Quantiles


Sampling Distribution of Mean: n = 500
Density
0
2
4
6
8
12
0.40
0.45
0.50
0.55
0.60
Sample mean


Q−Q Plot: n = 500
Sample Quantiles
0.40
0.50
0.60
−4
−2
0
2
4
Theoretical Quantiles


Mean and Median versus Sample Size (n)
Statistic value
0.46
0.48
0.50
Mean
Median
0
100
200
300
400
500
Sample size, n


Skewness versus Sample Size (n)
Skewness
0.0
0.2
0.4
0.6
0.8
0
100
200
300
400
500
Sample size, n


Kurtosis versus Sample Size (n)
Kurtosis
2.9
3.1
3.3
3.5
0
100
200
300
400
500
Sample size, n